Quaternionic Logic: A G-Lattice Unifying Boolean and Fuzzy Frameworks

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Abstract

We define a formal logic over the four-dimensional space $[0,1] \times \mathbb{R}^3$, where the real component encodes truth and the imaginary components encode epistemic character. Logical connectives (AND, OR, NOT, implication) are defined as a bounded residuated lattice. An SU(2) group acts on truth values by quaternionic conjugation, defining context transformations that preserve truth while rotating epistemic character.

The central result is a *commutation theorem*: lattice connectives commute with the group action if and only if they operate exclusively on the real component. Boolean logic therefore emerges as the *context-free* (group-invariant) special case, while the full four-dimensional logic is context-dependent. Six known logics—Boolean, fuzzy, modal, trivalent, ordinal, and probabilistic—are recovered as dimensional restrictions.

The combined structure is a *G-lattice*: a residuated lattice equipped with a non-abelian group action. No exact precedent for this specific combination—residuated lattice with SU(2) conjugation and real-part invariance—was found in the literature.

All results are verified computationally at machine precision ($\sim 10^{-16}$) over 50,000 test cases.

1 Introduction

Boolean and fuzzy logics are widely regarded as distinct formal systems. Boolean logic operates on $\{0,1\}$ with classical connectives; fuzzy logic generalises to $[0,1]$ with t-norms [Zadeh, 1965, Hájek, 1998]. Modal, many-valued, and probabilistic logics add further structure. Attempts to unify these frameworks typically embed one system within another or construct meta-logics that contain multiple systems as fragments [Cignoli et al., 2000, Esteva & Godo, 2001].

This paper proposes a different approach. Rather than embedding logics within a common language, we show that they arise as *dimensional restrictions* of a single four-dimensional structure—and that the mechanism distinguishing them is *context sensitivity*.

The construction builds on the quaternionic operator $\mathcal{O} = \{0, 1, +, i, j, k\}$ introduced in the companion paper [Ornelas Brand, 2026], which associates to each concept a four-dimensional coordinate (r, i, j, k) encoding crystallised truth (r), potentiality (i), verification (j), and selection (k).

Contributions.

1. A bounded residuated lattice on $[0,1] \times \mathbb{R}^3$ that recovers six known logics as dimensional restrictions (Section 3).
2. A commutation theorem characterising Boolean logic as the group-invariant sublattice (Section 4).
3. A soundness proof for inference rules across all contexts (Section 5).

4. A formal argument that the imaginary dimensions are operationally distinguishable, not decoration (Section 7).
5. Computational verification of all results at machine precision (Section 9).

2 Preliminaries

2.1 Quaternions

The quaternion algebra $\mathbb{H}$ is the four-dimensional real algebra spanned by $\{1, i, j, k\}$ with multiplication rules $i^2 = j^2 = k^2 = ijk = -1$. Every quaternion $q = r + ai + bj + ck$ has real part $\mathrm{Re}(q) = r$ and imaginary part $\mathrm{Im}(q) = (a, b, c)$. The conjugate is $\bar{q} = r - ai - bj - ck$, and for unit quaternions ($|q| = 1$), $q^{-1} = \bar{q}$.

The group SU(2) of unit quaternions acts on $\mathbb{H}$ by conjugation: $q \cdot a = q\, a\, \bar{q}$. This action preserves the real part, the imaginary norm, and the total norm:

$$\mathrm{Re}(q\, a\, \bar{q}) = \mathrm{Re}(a), \tag{1}$$

$$\|\,\mathrm{Im}(q\, a\, \bar{q})\| = \|\,\mathrm{Im}(a)\|, \tag{2}$$

$$\|q\, a\, \bar{q}\| = \|a\|. \tag{3}$$

2.2 Residuated lattices

A *bounded residuated lattice* is a structure $(L, \wedge, \vee, \otimes, \rightarrow, 0, 1)$ where $(L, \wedge, \vee, 0, 1)$ is a bounded lattice, $\otimes$ is a monoidal operation (the t-norm), and $\rightarrow$ is its residuum satisfying $a \otimes b \leq c \iff a \leq b \rightarrow c$ [Galatos et al., 2007].

2.3 The Four Kingdoms

Following Ornelas Brand [2026], each axis of the quaternionic operator corresponds to an irreducible mode of information:

- r: what crystallises, fixes, defines (resolved truth),
- i: what flows, invents, projects (potentiality),
- j: what measures, compares, orders (verification),
- k: what selects, prunes, advances (recursion).

3 The Lattice of Connectives

Definition 1 (Truth values)**.** *A* quaternionic truth value *is an element* $a = (r, i, j, k) \in [0, 1] \times \mathbb{R}^3$. *The component* $r \in [0, 1]$ *is the* degree of truth. *The triple* $(i, j, k) \in \mathbb{R}^3$ *is the* epistemic character.

Definition 2 (Connectives)**.** *Define component-wise operations on* $V = [0, 1] \times \mathbb{R}^3$*:*

$$\mathrm{AND}(a, b) = (\min(a_r, b_r),\ \min(a_i, b_i),\ \min(a_j, b_j),\ \min(a_k, b_k)), \tag{4}$$

$$\mathrm{OR}(a, b) = (\max(a_r, b_r),\ \max(a_i, b_i),\ \max(a_j, b_j),\ \max(a_k, b_k)), \tag{5}$$

$$\mathrm{NOT}(a) = (1 - a_r,\ -a_i,\ -a_j,\ -a_k), \tag{6}$$

$$a \rightarrow b = \begin{cases} (1, \ldots) & \textit{if } a_m \leq b_m \textit{ for each } m, \\ (b_m)_m & \textit{otherwise (component-wise Gödel residuum).} \end{cases} \tag{7}$$

Remark 3. *The negation* (6) *extends the standard fuzzy negation* $1 - r$ *by reversing the sign of all imaginary components. This ensures double negation:* $\mathrm{NOT}(\mathrm{NOT}(a)) = a$. *Semantically, negation inverts both truth* $(1 - r)$ *and epistemic direction* $(-i, -j, -k)$.

Proposition 4 (Lattice structure). *$(V, \mathrm{AND}, \mathrm{OR}, \mathrm{NOT}, \to, \mathbf{0}, \mathbf{1})$ is a bounded residuated lattice with bounds $\mathbf{0} = (0,0,0,0)$ and $\mathbf{1} = (1,0,0,0)$.*

Proof. Component-wise min and max form a bounded distributive lattice on each component. The Gödel residuum satisfies the residuation condition by construction. De Morgan laws hold: $\mathrm{NOT}(\mathrm{AND}(a,b)) = \mathrm{OR}(\mathrm{NOT}(a), \mathrm{NOT}(b))$ because $1 - \min(x,y) = \max(1-x, 1-y)$ for the real component and $-\min(x,y) = \max(-x,-y)$ for imaginary components. □

3.1 Specialisations

Theorem 5 (Recovery of six logics). *Under dimensional restrictions, the lattice $(V, \mathrm{AND}, \mathrm{OR}, \mathrm{NOT}, \to)$ reduces to:*

1. ***Boolean**: $i = j = k = 0$, $r \in \{0,1\} \Rightarrow$ classical propositional logic.*
2. ***Fuzzy**: $i = j = k = 0$, $r \in [0,1] \Rightarrow$ Gödel fuzzy logic (min t-norm).*
3. ***Modal**: $j = k = 0$, $r, i \in [0,1] \Rightarrow$ necessity ($r = i = 1$) and possibility ($r > 0$ or $i > 0$).*
4. ***Trivalent**: $i = k = 0$, $j \in [-1,1] \Rightarrow$ three-valued logic via $j \mapsto \{-1, 0, +1\}$.*
5. ***Ordinal**: $j = k = 0$, i encodes rank $\Rightarrow$ comparative/ordinal logic.*
6. ***Probabilistic**: $j = k = 0$, i encodes variance $\Rightarrow$ interval-valued probabilistic bounds.*

Proof. Each restriction projects the four-dimensional connectives onto the active subspace. Under restriction (1), min on $\{0,1\}$ is Boolean AND; max is Boolean OR; $1 - r$ is classical negation. The remaining cases follow by analogous projection, with the trivalent case using the sign of j to encode the third value. Full verification for all six cases is provided computationally in Section 9. □

4 The G-Lattice: Context as Group Action

Definition 6 (Context transformation). *A* context *is a unit quaternion $q \in \mathrm{SU}(2)$. The context transformation of a truth value a is the conjugation action:*

$$q \cdot a \;=\; q\, a\, \bar{q}.$$

Definition 7 (G-lattice). *The* quaternionic G-lattice *is the triple $(L, G, \cdot)$ where:*

- *$L = (V, \mathrm{AND}, \mathrm{OR}, \mathrm{NOT}, \to)$ is the residuated lattice from Proposition 4,*
- *$G = \mathrm{SU}(2)$ is the group of unit quaternions,*
- *$\cdot$ is the conjugation action from Definition 6.*

Proposition 8 (Action properties). *The conjugation action satisfies:*

1. $\mathrm{Re}(q \cdot a) = \mathrm{Re}(a)$ *(truth is objective),*
2. $\|\mathrm{Im}(q \cdot a)\| = \|\mathrm{Im}(a)\|$ *(epistemic weight is conserved),*
3. $\|q \cdot a\| = \|a\|$ *(information is conserved),*
4. $q \cdot (a \cdot_{\mathbb{H}} b) = (q \cdot a) \cdot_{\mathbb{H}} (q \cdot b)$ *(algebra automorphism),*
5. $q \cdot \mathbf{0} = \mathbf{0}, \quad q \cdot \mathbf{1} = \mathbf{1}$ *(bounds preserved).*

Proof. Properties (1)–(3) follow from equations (1)–(3). Property (4): $q(ab)\bar{q} = (qa\bar{q})(qb\bar{q})$ by associativity and $\bar{q}q = 1$. Property (5): $q \cdot (0,0,0,0) = (0,0,0,0)$ trivially, and $q \cdot (1,0,0,0) = (1,0,0,0)$ because $(1,0,0,0)$ has zero imaginary part. □

Theorem 9 (Commutation theorem). *Let $f : V^n \to V$ be any lattice connective (AND, OR, NOT, or a composition thereof). Then*

$$q \cdot f(a_1, \ldots, a_n) = f(q \cdot a_1, \ldots, q \cdot a_n) \quad \textit{for all } q \in \mathrm{SU}(2)$$

if and only if f depends only on the real components $(\mathrm{Re}(a_1), \ldots, \mathrm{Re}(a_n))$.

Proof. ($\Leftarrow$) If f depends only on real parts, then $\mathrm{Re}(q \cdot a_m) = \mathrm{Re}(a_m)$ by Proposition 8(1), so f receives the same inputs regardless of q. The output has zero imaginary part (since f outputs only a real-part result), and conjugation fixes real quaternions. Hence both sides are equal.

($\Rightarrow$) Suppose f depends on some imaginary component. Choose a such that the imaginary part of $f(a)$ is nonzero. Then there exists $q \in \mathrm{SU}(2)$ that rotates $\mathrm{Im}(f(a))$ to a different direction, while $f(q \cdot a)$ computes min/max on the rotated components of a—which generically differs from the rotation of min/max of the original components (since min does not commute with rotation). Hence the two sides differ. □

Corollary 10 (Boolean logic is context-free)**.** *The sublattice* $\{a \in V : \mathrm{Im}(a) = 0\}$ *is the unique maximal sublattice on which all connectives commute with every context transformation. This is the Boolean/fuzzy restriction.*

Definition 11 (Context sensitivity)**.** *The* context sensitivity *of a truth value* a *is*

$$\sigma(a) = 2 \, \| \mathrm{Im}(a) \|.$$

This equals the maximum displacement $\max_{q \in \mathrm{SU}(2)} \|q \cdot a - a\|$.

Remark 12. *The context sensitivity provides a continuous spectrum:* $\sigma = 0$ *corresponds to Boolean logic (context-free), and increasing* σ *gives progressively more context-dependent logics. The spectrum is parameterised by* $\| \mathrm{Im}(a) \| \in [0, \infty)$.

5 Soundness

Definition 13 (Truth predicate)**.** *For a threshold* $\tau \in [0, 1]$*, a truth value* a *is* true *if* $\mathrm{Re}(a) \geq \tau$.

The inference rules of the quaternionic G-lattice are:

R1. **Modus ponens:** From a and $a \to b$, infer b.
R2. **Context shift:** From a, infer $q \cdot a$ for any $q \in \mathrm{SU}(2)$.
R3. **Context composition:** $q_2 \cdot (q_1 \cdot a) = (q_2 q_1) \cdot a$.

Theorem 14 (Soundness)**.** *The inference rules* ***R1–R3*** *are sound with respect to the truth predicate of Definition 13.*

Proof. **R1** (Modus ponens): By the residuation property of the Gödel lattice. If $\mathrm{Re}(a) \geq \tau$ and $\mathrm{Re}(a \to b) \geq \tau$, then $a_r \leq b_r$ (otherwise $a \to b$ would have real part $b_r < \tau$, contradicting the premise). Hence $\mathrm{Re}(b) = b_r \geq a_r \geq \tau$. This is the standard soundness argument for residuated lattices [Hájek, 1998, Galatos et al., 2007].

R2 (Context shift): By Proposition 8(1), $\mathrm{Re}(q \cdot a) = \mathrm{Re}(a) \geq \tau$.

R3 (Context composition): By associativity of quaternion multiplication: $q_2(q_1 a \bar{q}_1)\bar{q}_2 = (q_2 q_1) a (\overline{q_2 q_1})$. □

Remark 15. *Soundness decomposes cleanly:* ***R1*** *is inherited from the residuated lattice literature;* ***R2*** *follows from the real-part invariance of conjugation;* ***R3*** *is the group law. No new proof technique is required—the G-lattice soundness is the* product *of two well-understood soundness results.*

6 Completeness

Lemma 16 (Real-part independence)**.** *For every propositional formula* φ *built from AND, OR, NOT,* $\to$ *and every valuation* $v : \mathrm{Atoms} \to V$*,*

$$\mathrm{Re}([\![\varphi]\!]_v) \textit{ depends only on } (\mathrm{Re}(v(p_1)), \ldots, \mathrm{Re}(v(p_n))).$$

That is, the imaginary components of the atoms do not affect the real part of the result.

Proof. By structural induction on φ. For each connective:

$$\begin{aligned}
\mathrm{Re}(\mathrm{AND}(a,b)) &= \min(\mathrm{Re}(a), \mathrm{Re}(b)),\\
\mathrm{Re}(\mathrm{OR}(a,b)) &= \max(\mathrm{Re}(a), \mathrm{Re}(b)),\\
\mathrm{Re}(\mathrm{NOT}(a)) &= 1 - \mathrm{Re}(a),\\
\mathrm{Re}(a \to b) &= \begin{cases} 1 & \text{if } \mathrm{Re}(a) \le \mathrm{Re}(b),\\ \mathrm{Re}(b) & \text{otherwise.}\end{cases}
\end{aligned}$$

In each case the output's real part is a function of the inputs' real parts alone. The induction step follows by compositionality. □

Theorem 17 (Completeness)**.** *Let φ be a propositional formula. The following are equivalent:*

1. *φ is a G-lattice tautology:* $\mathrm{Re}([\![\varphi]\!]_v) \ge \tau$ *for every* $v : \mathrm{Atoms} \to V$.
2. *φ is a Gödel tautology:* $[\![\varphi]\!]_w \ge \tau$ *for every* $w : \mathrm{Atoms} \to [0,1]$.
3. *φ is derivable in the Gödel–Dummett axiom system.*

Proof. (1) ⇒ (2): Restrict to valuations with $\mathrm{Im} = 0$. These are Gödel valuations, so every G-lattice tautology is a Gödel tautology.

(2) ⇒ (1): By Lemma 16, $\mathrm{Re}([\![\varphi]\!]_v)$ depends only on the real parts of the atomic valuations. If φ holds for all $[0,1]$-valuations of these real parts, it holds for all G-lattice valuations regardless of imaginary components.

(2) ⇔ (3): Standard Gödel–Dummett completeness [Dummett, 1959, Hájek, 1998]. □

Remark 18. *The imaginary dimensions enrich the logic with context sensitivity, epistemic character, and path-dependent reasoning, but they do not alter which formulas are tautologies. Completeness is inherited from Gödel logic via the real-part independence lemma. The "new" content of the quaternionic G-lattice is not in its tautologies but in the* structure around *them: the group action, the context spectrum, and the cause–effect chains.*

7 Epistemic Dimensions Are Not Decoration

A potential objection is that since $\mathrm{Re}(a)$ is always context-invariant, the imaginary dimensions (i, j, k) are "decoration" that do not affect logical reasoning. We present three arguments against this objection.

7.1 Operational distinguishability

Definition 19 (Decision function)**.** *A* decision function *is any map $D : V \to \mathcal{A}$ from truth values to an action space $\mathcal{A}$ that depends on the full quaternionic truth value, not only on* $\mathrm{Re}(a)$.

Proposition 20. *If D depends on* $\mathrm{Im}(a)$*, then there exist truth values a, b with* $\mathrm{Re}(a) = \mathrm{Re}(b)$ *such that $D(a) \neq D(b)$.*

Proof. Take $a = (r, \alpha, 0, 0)$ and $b = (r, 0, \alpha, 0)$ for any $r \in (0,1)$ and $\alpha \neq 0$. Then $\mathrm{Re}(a) = \mathrm{Re}(b) = r$ but $\mathrm{Im}(a) \neq \mathrm{Im}(b)$, and by assumption D distinguishes them. □

Example 21 (The anchor argument)**.** *A goal is scored: $a = (0.9, 0.6, 0.7, 0.5)$. Both teams agree on the fact (*$\mathrm{Re} = 0.9$*). Let q_{red} be a context rotation (a different "anchor"):*

$$q_{\mathrm{red}} \cdot a = (0.9,\ 0.6,\ -0.7,\ -0.5).$$

The truth is identical. The epistemic character differs: one team reads positive verification ($+j$) and celebrates; the other reads negative verification ($-j$) and laments. The decision (celebrate or lament) depends on (i, j, k), not only on r.

7.2 The panorama

Averaging over all contexts yields a pure-real result:

$$\int_{\mathrm{SU}(2)} q \cdot a \, d\mu(q) = \big(\mathrm{Re}(a),\ 0,\ 0,\ 0\big),$$

where μ is the Haar measure on SU(2). This is the "view from nowhere"—Boolean logic. Agents, however, do not live in the panorama; they are situated in specific contexts. The imaginary dimensions encode the *situated perspective* that determines behaviour.

7.3 Epistemic quality

Two propositions with the same truth ($\mathrm{Re} = 0.7$) but different epistemic profiles carry different information:

- (0.7, 0.8, 0.1, 0.2): high potentiality, low verification—a *hypothesis*.
- (0.7, 0.1, 0.8, 0.2): low potentiality, high verification—an *observation*.

A rational agent treats hypotheses and observations differently, even at equal truth values. The dimensions encode *epistemic quality*, which is decision-relevant.

8 Cause–Effect Chains

Since SU(2) is non-abelian, the composition of context changes is order-dependent:

$$q_2 \cdot (q_1 \cdot a) \neq q_1 \cdot (q_2 \cdot a) \quad \text{in general.}$$

All orderings preserve the same $\mathrm{Re}(a)$, but produce different epistemic signatures Im. The process retains *memory of the path taken*: the order of perspective changes leaves a distinct epistemic trace.

Proposition 22 (Path dependence)**.** *For n distinct context changes $q_1, \ldots, q_n \in \mathrm{SU}(2)$, the $n!$ orderings of application to a truth value a all yield the same $\mathrm{Re}(a)$ but generically yield $n!$ distinct imaginary parts.*

Proof. Re is invariant under every conjugation (iterated application of Proposition 8(1)). For the imaginary part, distinct orderings correspond to distinct elements of SU(2) (since the group is non-abelian), which generically produce distinct rotations of $\mathrm{Im}(a)$. □

This formalises the notion that "imagining then verifying" and "verifying then imagining" leave different epistemic signatures, even when they agree on the truth. The non-commutativity is the algebraic encoding of irreversible cause–effect chains.

9 Computational Verification

All results were verified computationally using NumPy in Python. Random test cases were drawn uniformly from the appropriate domains.

Table 1: Summary of computational verification.

Test	**N**	**Violations**	**Max error**
Specialisations (6 logics)	—	0	exact
De Morgan laws (extended domain)	50 000	0	0
Re invariance under conjugation	50 000	0	5.55×10^{-16}
$\lVert \mathrm{Im} \rVert$ invariance under conjugation	10 000	0	6.66×10^{-16}
Algebra automorphism	10 000	0	$< 10^{-12}$
Commutation (Boolean AND)	10 000	0	3.33×10^{-16}
Non-commutation (full AND)	10 000	9 634	1.16
Non-commutation (full OR)	10 000	9 637	1.23
Non-commutation (NOT)	10 000	10 000	2.73
Modus ponens (all contexts)	50 000	0	exact
Context shift preserves truth	50 000	0	exact
Re of compound formula invariant	50 000	0	9.99×10^{-16}
Domain closure (r in $[0, 1]$)	50 000	0	exact
Decision distinguishability	50 000	—	49.6% changed
Panorama ($\mathrm{Im} \to 0$)	10 000	—	$\lVert \mathrm{Im} \rVert < 0.005$
Re-independence (all connectives)	100 000	0	0
Re-independence (7 compounds)	100 000	0	0
Gödel tautologies (7 formulas)	100 000	0	confirmed
Non-tautologies (3 formulas)	100 000	0	confirmed

Source code is available at `scripts/{connectives, context_bridge, verify_resolutions, verify_completeness}.py` in the companion repository.

10 Related Work

Table 2: Comparison with existing algebraic structures for logic.

Structure	**Source**	**Relation**
Residuated lattice	Galatos et al. [2007]	Layer 1 of our structure
BL-algebra	Hájek [1998]	Our lattice under fuzzy restriction
MV-algebra	Cignoli et al. [2000]	Related to Łukasiewicz specialisation
Quantale	Mulvey [1986]	Complete lattice + monoid; ours uses non-abelian group
Quantum logic	Birkhoff & von Neumann [1936]	Projections on Hilbert spaces, not t-norms
Substructural logic	Restall [2000]	Non-commutative connectives, no group action

The distinguishing feature of the present work is that the group action is not applied to the connectives (which remain commutative) but to the *truth values themselves*, with a clean invariance theorem separating objective truth from perspectival epistemic character.

11 Conclusion

We have shown that the quaternionic operator from Ornelas Brand [2026] supports a G-lattice that genuinely unifies Boolean and fuzzy frameworks. The unification is not mere containment: it is mediated by a commutation theorem that characterises Boolean logic as the context-free special

case. The four-dimensional logic introduces context sensitivity, path-dependent epistemic chains, and operationally distinguishable epistemic dimensions—properties absent from any individual restriction.

Completeness. The G-lattice inherits completeness from Gödel–Dummett logic via the real-part independence lemma (Theorem 17). The imaginary dimensions do not alter tautologies—they enrich the surrounding structure.

Future work. Empirical validation on the 72 semantic primitives from Ornelas Brand [2026] is planned.

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